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0625/63

October/November 2018

1 hour

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

This syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **12** printed pages and **4** blank pages.

- 1 A student is determining the density of modelling clay.

He is using the block shown in Fig. 1.1 and Fig. 1.2.

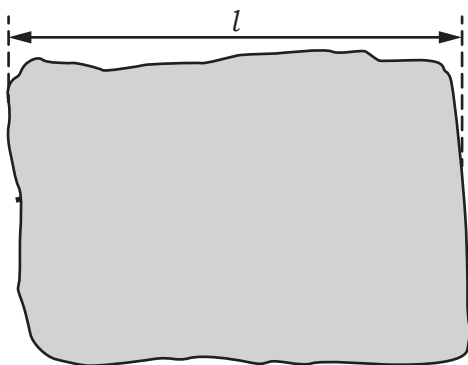


Fig. 1.1

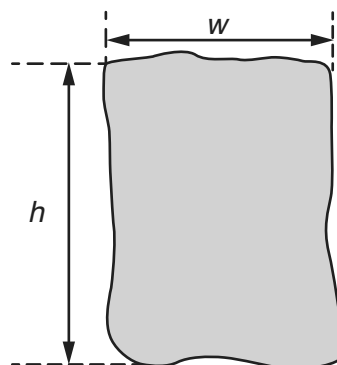


Fig. 1.2 (side view)

- (a) (i) Measure the dimensions of the block of modelling clay, as shown in Fig. 1.1 and Fig. 1.2.

length $l =$ cm

width $w =$ cm

height $h =$ cm
[1]

- (ii) Calculate the volume V_1 of the block, using your measurements from (a)(i) and the equation $V_1 = l \times w \times h$.

$V_1 =$ cm³ [1]

- (b) Suggest a possible source of inaccuracy in measuring the dimensions of the block and describe an improvement to the procedure that will produce more reliable measurements of the block.

suggestion

.....

improvement

.....

.....

[2]

- (c) The student suspends the piece of modelling clay from a forcemeter, as shown in Fig. 1.3.

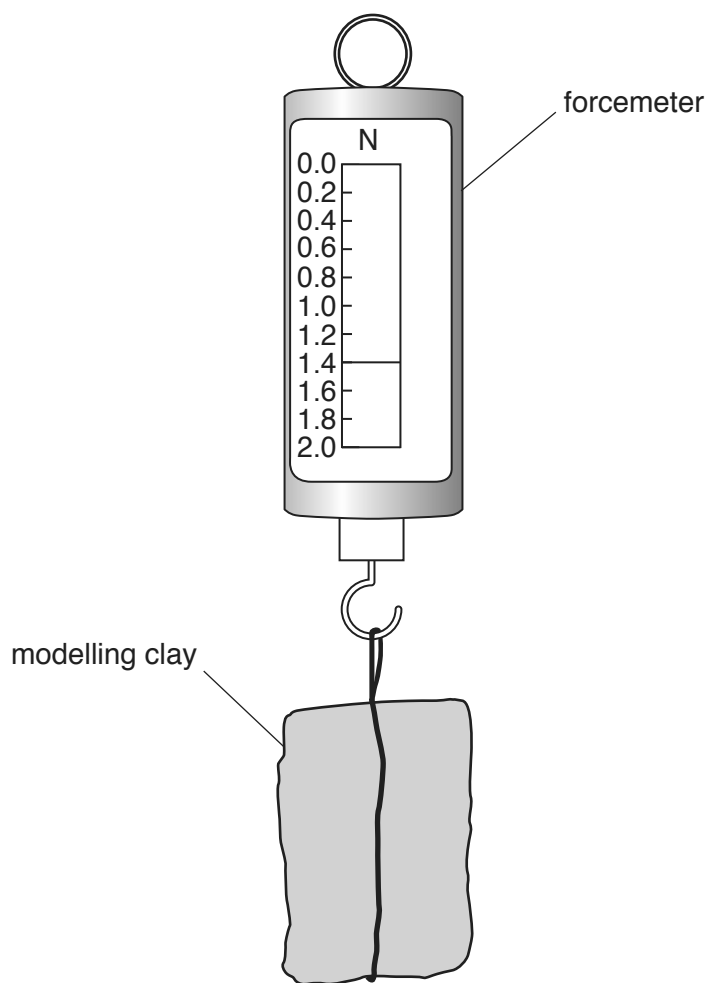


Fig. 1.3

Record the weight W of the block of modelling clay shown in Fig. 1.3.

$W = \dots\dots\dots$ N [1]

- (d) Calculate a value ρ_1 for the density of the modelling clay, using your results from (a)(ii) and (c) and the equation

$$\rho_1 = \frac{W \times k}{V_1},$$

where $k = 100 \text{ g/N}$.

$\rho_1 = \dots\dots\dots$ [2]

- (e) The student pours some water into a measuring cylinder, as shown in Fig. 1.4.

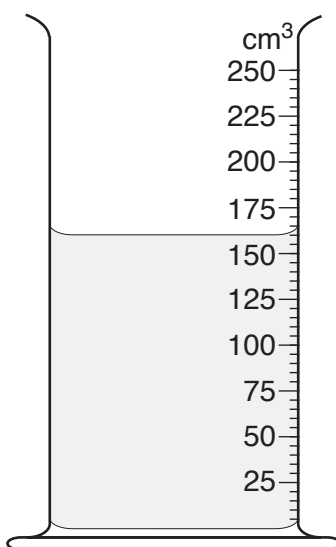


Fig. 1.4

- (i) Record the volume V_2 of the water in the measuring cylinder shown in Fig. 1.4.

$V_2 = \dots\dots\dots \text{cm}^3$ [1]

- (ii) Describe how a measuring cylinder is read to obtain an accurate value for the volume of water. You may draw a diagram to help you.

.....

.....

..... [1]

- (f) The student lowers the modelling clay into the water, as shown in Fig. 1.5.

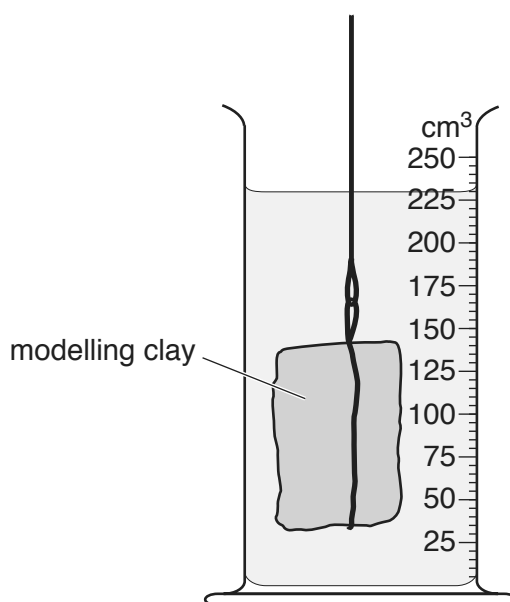


Fig. 1.5

- (i) • Record the new reading V_3 of the measuring cylinder in Fig. 1.5, with the block of modelling clay in the water.

$$V_3 = \dots\dots\dots \text{cm}^3$$

- Calculate another value ρ_2 for the density of modelling clay, using your value for V_3 , your readings from (c) and (e)(i) and the equation

$$\rho_2 = \frac{W \times k}{(V_3 - V_2)},$$

where $k = 100 \text{ g/N}$.

$$\rho_2 = \dots\dots\dots [1]$$

- (ii) Suggest which of ρ_1 or ρ_2 is likely to be the more accurate value for the density of the modelling clay.

Justify your answer by referring to the procedure.

.....

 [1]

[Total: 11]

- 2 Some students are determining the resistance per unit length of wire **X**.

They are using the circuit shown in Fig. 2.1.

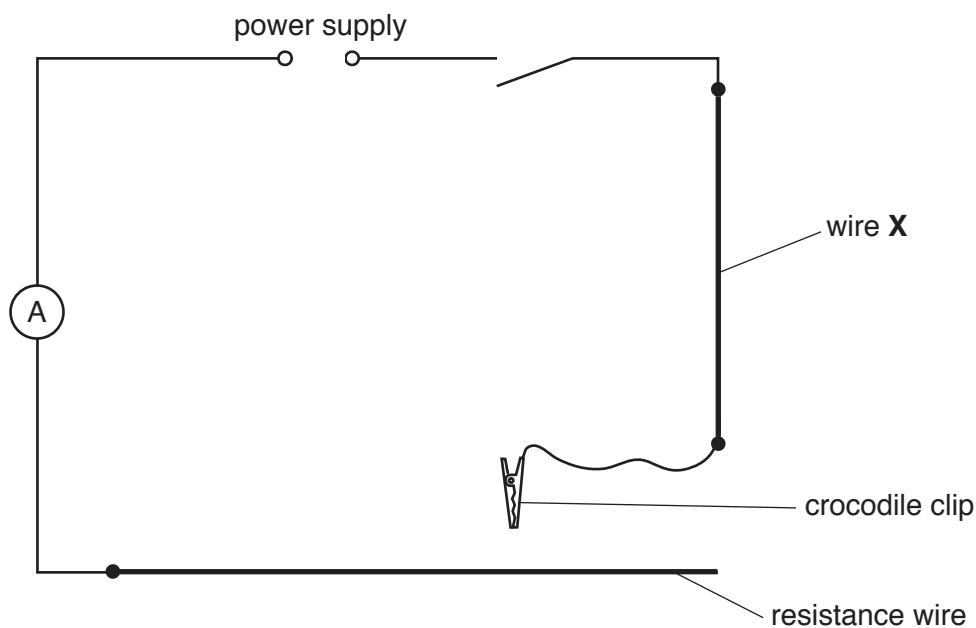


Fig. 2.1

The crocodile clip is connected to a length of the resistance wire. The current in the circuit and potential difference (p.d.) across wire **X** are measured.

- (a) On Fig. 2.1, draw the symbol for a voltmeter correctly connected to measure the p.d. across wire **X**. [1]
- (b) A student adjusts the position of the crocodile clip until the current in the circuit is 0.80 A.

In Table 2.1, record the value of the p.d. V across wire **X**, as shown in Fig. 2.2.

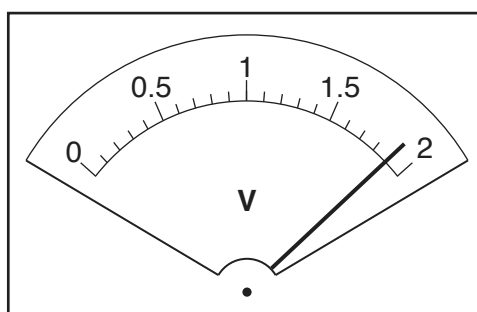


Fig. 2.2

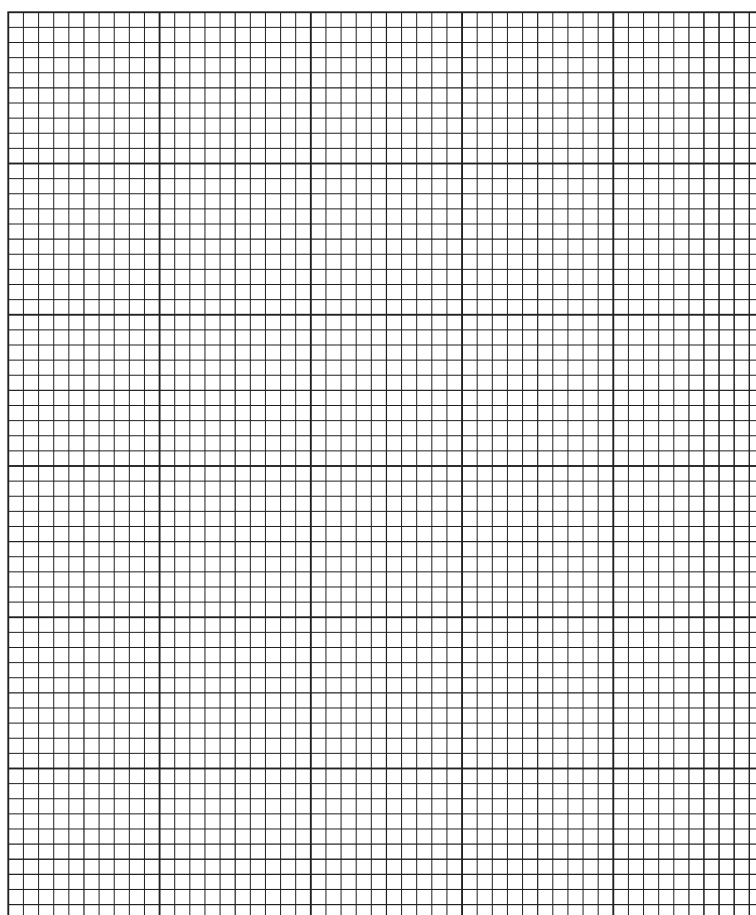
[1]

- (c) The student records the p.d. V for different currents, as shown in Table 2.1.

Table 2.1

I/A	V/V
0.80	
0.70	1.58
0.60	1.34
0.50	1.03
0.40	0.81

- (i) Plot a graph of V/V (y -axis) against I/A (x -axis).



[4]

- (ii) Determine the gradient G of the graph. Show clearly on the graph how you obtained the necessary information.

$G = \dots\dots\dots$ [1]

(iii) Fig. 2.3 shows wire **X**.

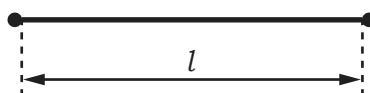


Fig. 2.3

- On Fig. 2.3, measure the length l .

$l =$ cm

- The wire in Fig. 2.3 is shown to $1/3^{\text{rd}}$ scale. Calculate the actual length L of wire **X**.

$L =$ cm
[1]

(iv) Calculate the resistance R of each centimetre of wire **X**, using the equation

$$R = \frac{G \times k}{L},$$

where $k = 1.0 \Omega \text{cm}$.

Give the value for R , to a suitable number of significant figures for this experiment.

$R =$ [2]

(d) A student notices that the resistance wire becomes very hot when the crocodile clip is connected to short lengths of the wire.

Suggest an improvement that would help to reduce this effect.

.....

 [1]

[Total: 11]

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- 3 A student is investigating the reflection of light by a plane mirror.

Her ray-trace sheet is shown in Fig. 3.1.

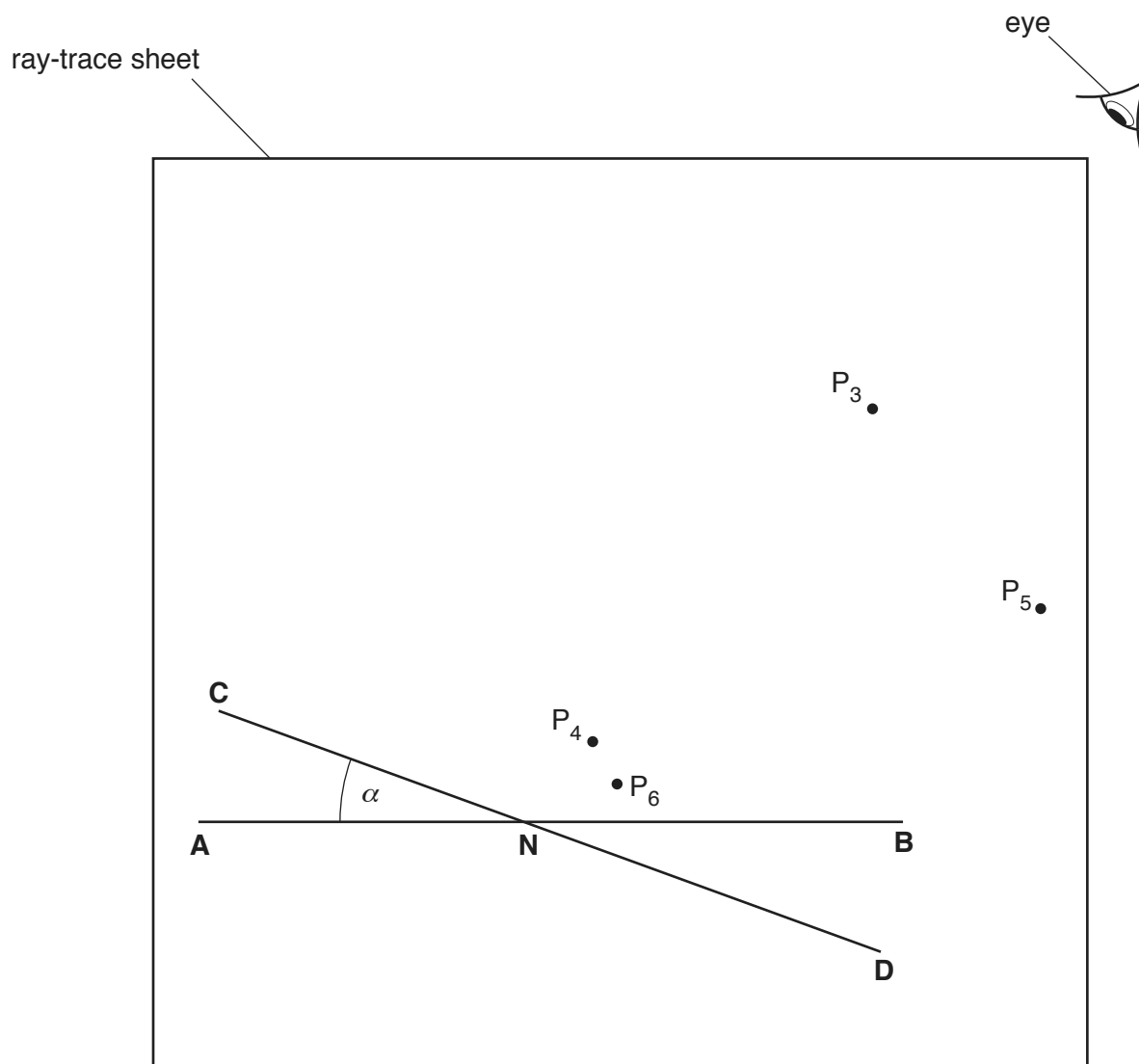


Fig. 3.1

- (a) (i) Measure and record in Table 3.1 the angle α between **AN** and **CN**, as shown in Fig. 3.1. [1]
- (ii) Draw a normal to **AB** at point **N**, towards the top of the page. Label the top point of this normal **L**. [1]
- (iii) Two pins P_1 and P_2 are placed on line **LN**, a suitable distance apart for ray tracing. On Fig. 3.1, label suitable positions for P_1 and P_2 . [1]

- (b) The student places the reflecting surface of a plane mirror on line **CD** and views the images of P_1 and P_2 from the direction indicated by the eye in Fig. 3.1.

She places two pins P_3 and P_4 so that the images of P_1 and P_2 , and the pins P_3 and P_4 , all appear exactly one behind the other.

- (i) Draw a line passing through P_3 and P_4 and reaching point **N**. [1]

- (ii) Measure the angle θ between this line and the normal **LN**.
Record this value in Table 3.1. [1]

Table 3.1

	$\alpha/^\circ$	$\theta/^\circ$
mirror on CD		
mirror on EF	35	

- (c) The student moves the mirror to a line **EF** at an angle $\alpha = 35^\circ$ to **AN**. Line **EF** is not shown in Fig. 3.1.
The student repeats the process with the mirror on **EF** and places pins P_5 and P_6 in line with the new images.

- (i) Draw a line passing through P_5 and P_6 and reaching point **N**. [1]

- (ii) Measure the angle θ between this new line and the normal **LN**.
Record this value in Table 3.1. [1]

- (d) A student suggests that when the mirror is moved, the **change in** θ should be equal to twice the **change in** α .

State whether your readings support this idea. Justify your answer with reference to the readings.

statement

justification

.....

.....

[2]

- (e) Suggest two precautions that you would take to ensure accurate results from this type of experiment.

1

.....

2

.....

[2]

[Total: 11]

- 4 A student is investigating the conduction of thermal energy by metals.

Plan an experiment to compare the rates at which different metals conduct thermal energy.

The apparatus available includes:

strips of different metals, shaped as shown in Fig. 4.1
a test-tube in a clamp stand
a beaker
a supply of cold water
a supply of hot water.

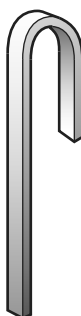


Fig. 4.1

The shorter section of each strip of metal can fit inside a test-tube.

Write a plan for the experiment.

You should:

- list any additional apparatus needed
- draw a labelled diagram of how the apparatus will be arranged
- explain briefly how you will carry out the experiment
- explain how the metals will be compared
- state the precautions which should be taken to obtain reliable results.

[Total: 7]

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